10.5.1 Analog and Digital Signals

There are 2 different types of signals used in Electronics. They are (i) Analog signals and (ii) Digital signals. An analog signal is a continuously varying voltage or current with respect to time. Such signals are employed in rectifying circuits and transistor amplifier circuits.

Digital signals are signals which contain only discrete values of voltages. Digital signals need two states: switch ON and OFF. ON is considered as one state and OFF is considered as the other state. It can also be defined as high (ON) or low (OFF) state, closed (ON) or open (OFF). These high and low states are defined using binary numbers 1 or 0 in Boolean Algebra. The state 1 represents the terms: circuit on, high voltage, a closed switch. Similarly a 0 state represents circuit off, low voltage or an open switch.

Positive and Negative Logic

In digital systems, there exists two voltage levels: 5V (high) and 0V (low). In a positive logic system; a binary 1 stands for 5V and 0 stands for 0V while in negative logic system, 1 stands for 0V and 0 stands for 5V as shown in Figure 10.40.

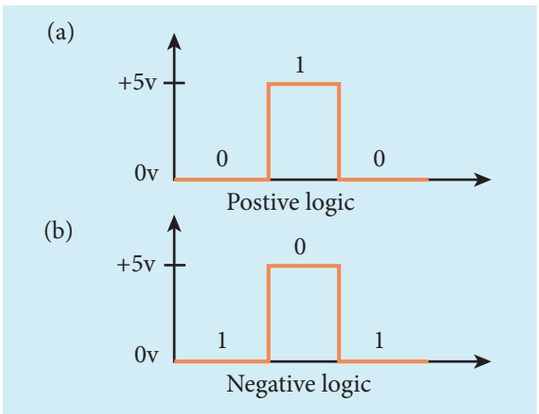


Figure 10.40 (a) Positive logic (b) Negative logic

10.5.2 Logic gates

A logic gate is an electronic circuit whose function is based on digital signals. They are binary in nature. The logic gates are considered as the basic building blocks of most of the digital systems. They have one output with one or more inputs. There are three types of basic logic gates: AND, OR, and NOT. The other logic gates are Ex-OR, NAND, and NOR. They can be constructed from the basic logic gates.

Digital electronics deals with logical operations. The variables are called logical variables. The operators like logical addition (+) and logical multiplication (·) are called logical operators. When the logical operators (+, .) operate on logical variables (A, B), they give logical constant (Y). The equation that represents this operation is called logical statement.

For example,

Logical operator: +

Logical variable: A, B

Logical constant: Y

Logical statement: $Y = A + B$

The possible combinations of inputs and the corresponding output are given in the form of a table called truth table. The circuits which perform the basic logical operations such as logical addition, multiplication and inversion are discussed below.

AND gate

Circuit symbol

The circuit symbol of a two input AND gate is shown in Figure 10.41(a). A and B are inputs and Y is the output. It is a logic gate and hence A, B, and Y can have the value of either 1 or 0.

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